INCH-POUND
MIL-PRF-3098/92F
6 March 2013
SUPERSEDING
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16 January 2006

PERFORMANCE SPECIFICATION SHEET

# CRYSTAL UNIT, QUARTZ, CR116/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 125 MHz; fifth mechanical overtone; noncontrolled; series resonance.

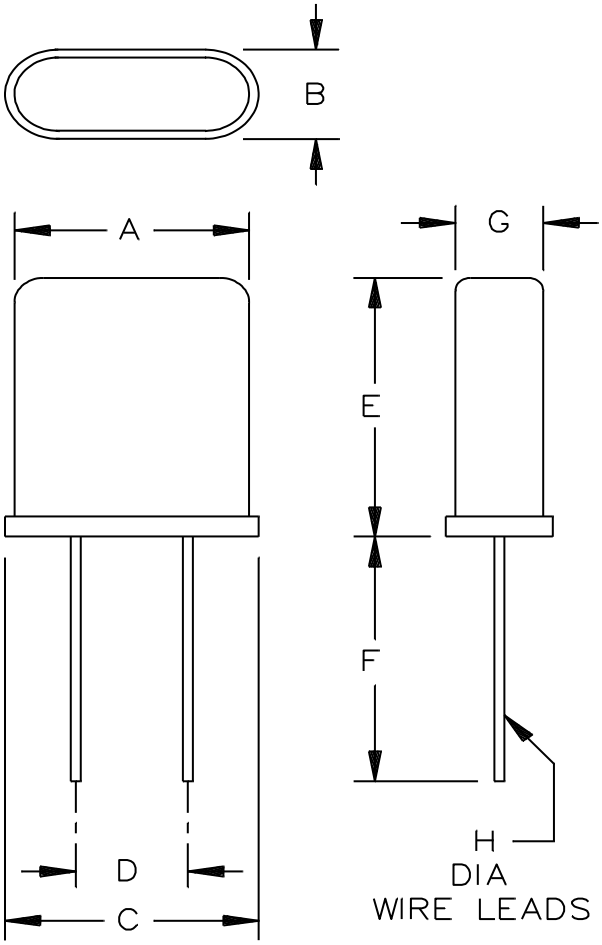



FIGURE 1. Crystal unit - CR116/U.



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| Ltr | Inches | | mm | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | --- | .402 | --- | 10.21 |
| B | --- | .183 | --- | 4.65 |
| C | --- | .435 | --- | 11.05 |
| D | .184 | .200 | 4.67 | 5.08 |
| E | --- | .530 | --- | 13.46 |
| F | .50 | --- | 12.7 | --- |
| G | --- | .150 | --- | 3.81 |
| H | .015 | .019 | 0.38 | 0.48 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR116/U, crystal unit - Continued.



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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 MHz to 125 MHz, inclusive.

Capacitance, shunt: 4.5 pF, maximum.

Mode of oscillation: Fifth mechanical overtone.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Calibration values:

Resistance: 40 ohms.

Resistor voltage drop: 0.28 volt.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Frequency tolerance: ±25 ppm.

Equivalent resistance: 60 ohms, maximum.

Unwanted modes: The minimum unwanted mode effective resistance shall be at least three times that of the main mode, or 120 ohms, whichever is greater.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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